Why Atoms Bond

Checkpoint Answer Sheet

Student name: ________________________________ Date: ________________

Complete independently

Use complete sentences and label every charge. Submit only this completed checkpoint sheet to U3 L1 Checkpoint Submission.

Scenario

Magnesium has two valence electrons. Oxygen has six valence electrons. A stable ionic arrangement forms when magnesium transfers two electrons to oxygen.

1. State the charge formed by magnesium and the charge formed by oxygen. Explain how each charge forms.

2. Draw a labeled before-and-after particle model. Show Mg and O before transfer, two transferred electrons, Mg2+ and O2- after transfer, and an arrow labeled electrostatic attraction.

BEFORE: neutral atoms	AFTER: ions and attraction

3. Explain why the ions remain together after transfer. Use the terms opposite charges, electrostatic attraction, and lower-energy arrangement.

4. Evidence statement: Identify one feature of your model that proves a bond-forming attraction is present.